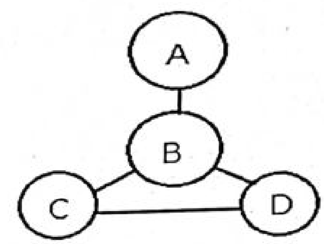
JMO – 2023 (21^{ST} Jan-2024)

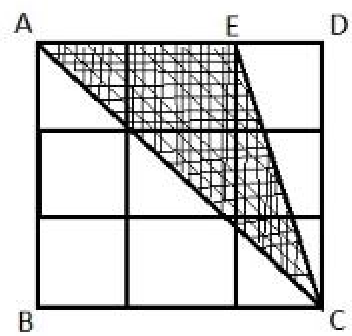
Answer all questions. All questions carry equal marks.

Full Marks 100

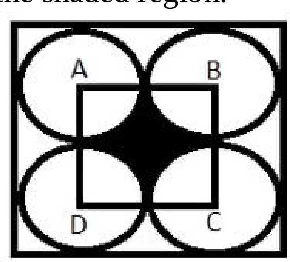
1. The product of the ages of 3 students is 210 and sum of their ages is 18, Find the ages of each students.
2. How many ways are there to colour the circle of the figure given, by using only 3 different colours so that no two circles joined by a line have the same colour.



3. Evaluate: $\left(\frac{1}{2}+\frac{1}{3}+...+\frac{1}{10}\right)+\left(\frac{2}{3}+\frac{2}{4}+...+\frac{2}{10}\right)+\left(\frac{3}{4}+\frac{3}{5}+...+\frac{3}{10}\right)+,,,+\left(\frac{8}{9}+\frac{8}{10}\right)+\frac{9}{10}$.
4. In the given figure, a square of side 3 cm is divided into 9 small squares of side 1 cm. Find the area of the shaded portion as shown in the figure.



5. Find the value of a, b, c, d, e and f, if $999\times abc = def132$, where abc is a three digit number and $def132$ is a six digit number.
6. In the given figure, in a square of area 16 sq cm, 4 small circles of each of same area are drawn and a portion is shaded. Find the area of the shaded region.



7. Samalkar walked from his home towards railway station at 60 m/min. At the same time, his brother returned from railway station at 40 m/min. They met 100 m away from the middle of the whole journey. How far is the railway station from their home?
8. In an examination there are 30 questions. For a correct answer one gets 5 mark and for a wrong answer 2 marks are deducted. If one gets total of 122 marks, then how many are correct and how many are wrong answered?

9. From a group of boys and girls, 15 girls left the group then the ratio of boys to girls becomes 2:1. After that from the same group 45 boys left then the ratio of the boys to girls become 1:5. How many girls were there in the beginning?
10. Find the largest natural number ‘n’ for which $n^{6072} < 2024^{2024}$.
11. What digit does each letter represent?

$$\begin{array}{cccccc} A & B & C & D & E & F \\ \times & & & & & F \\ \hline G & G & G & G & G & G \end{array}$$

12. A cricket ball dashed on to the window pan in the staff room and shattered it. Four probable culprits were called to the school discipline incharge for investigation. The answered as follows:
    Aryan: Deepak broke it,
    Bibhu: Deepak broke it.
    Chandan: I didn’t do it.
    Deepak: Bibhu is lying.
    Only one suspect told the truth. Who was the culprit?
13. At what time between 5 and 6 will the hands of a clock be at right angle?
14. Divide 127 into 4 parts such that if the 1^{st} part is increased by 18, 2^{nd} part is decreased by 5, 3^{rd} part is multiplied by 6 and 4^{th} part is divided by $2\frac{1}{2}$, then the results are same. Find these 4 numbers.
15. In the given table, in which row and in which column the number 2024 appear?

A	B	C	D	E
	8	6	4	2
10	12	14	16	
	24	22	20	18
26	28	30	32	
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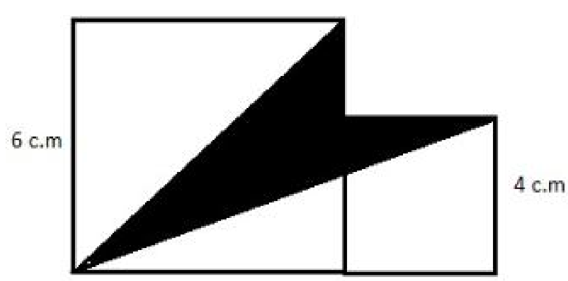
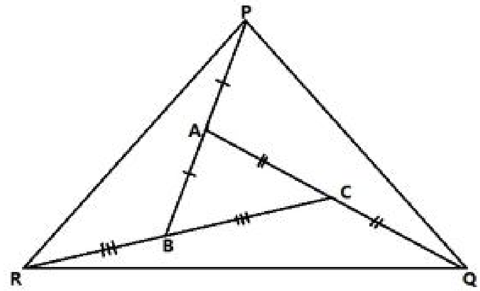
16. Find area of the shaded part in the given figure when lengths of the given 2 squares are given to be 6 cm and 4 cm.

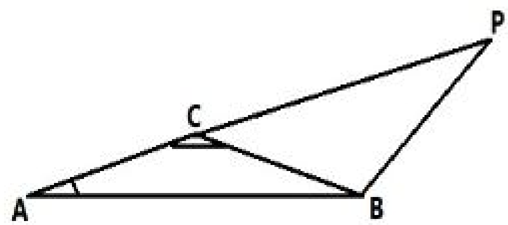


17. Divide 306 balls between A, B, C such that $\frac{1}{4}^{th}$ balls of A = $\frac{3}{8}^{th}$ balls of B = $\frac{5}{12}^{th}$ balls of C

18. In ΔABC, $\overline{BA}$, $\overline{AC}$ and $\overline{CB}$ are produced to P, Q and R respectively such that $\overline{AB} = \overline{AP}$, $\overline{AC} = \overline{CQ}$ and $\overline{CB} = \overline{BR}$. If area of ΔPQR is 210 sq cm. Find the area of ΔABC.



19. In ΔABC, $\angle A = 40^0$, $\angle ACB = 120^0$. AC is produced to P such that AP = AC+2BC. Find $\angle ABP$.



20. If $a = \cfrac{1}{b + \cfrac{1}{c + \cfrac{1}{d + \cfrac{1}{e}}}} = \dfrac{361}{305.}$, find the value of a+b+c+d+e.

Note: here Question is misprinted.

Exact question is: $a + \cfrac{1}{b + \cfrac{1}{c + \cfrac{1}{d + \cfrac{1}{e}}}} = \dfrac{361}{305.}$, find the value of a+b+c+d+e.

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